ECEN 428/722 - Lab Report

Lab Number: 4

Lab Title: Discrete Fourier Transform on FPGAs

Section Number: 602

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Objectives:

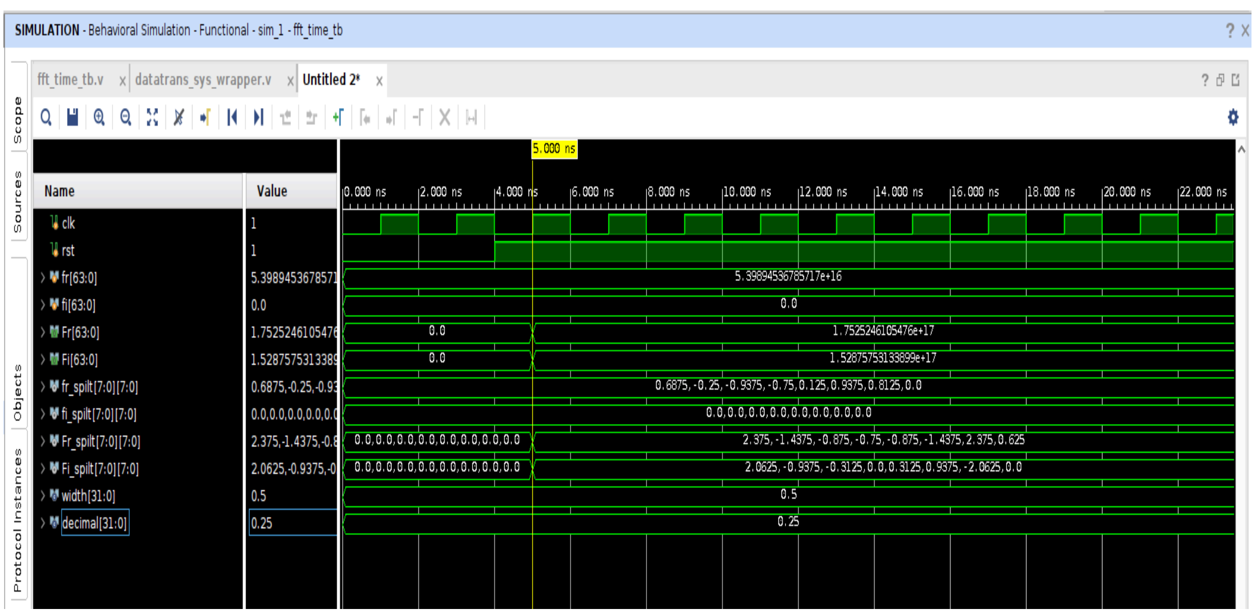
Run and justify a **behavioral simulation** of the radix-2 **decimation-in-time** (DIT) FFT design. The lab uses 8 inputs and 8 outputs (each 8-bit signed Q4.4) organized in **3 stages**; your pre-lab deliverables are screenshots of the simulation and a brief correctness explanation.

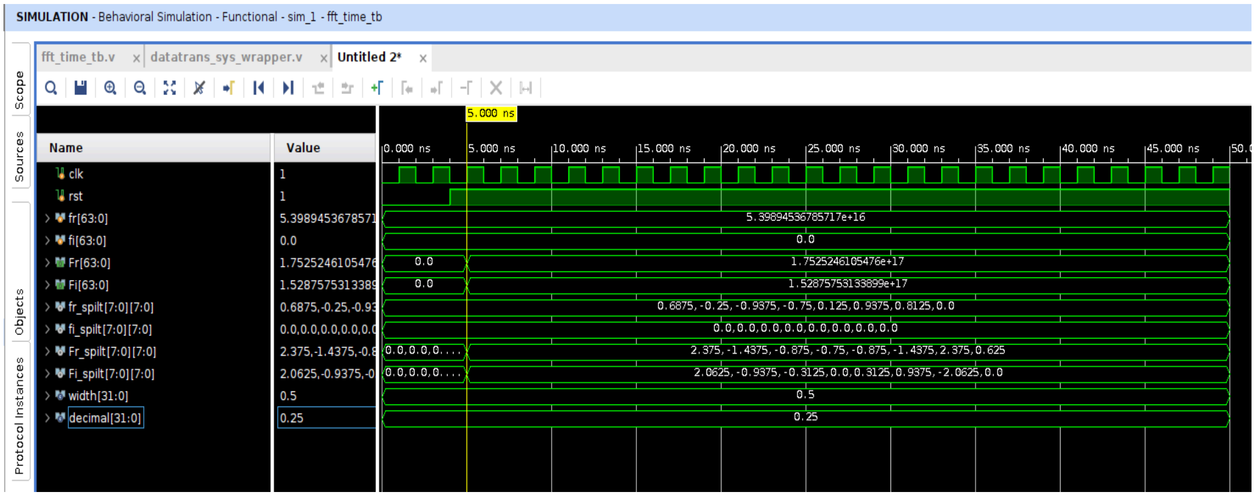
Setup:

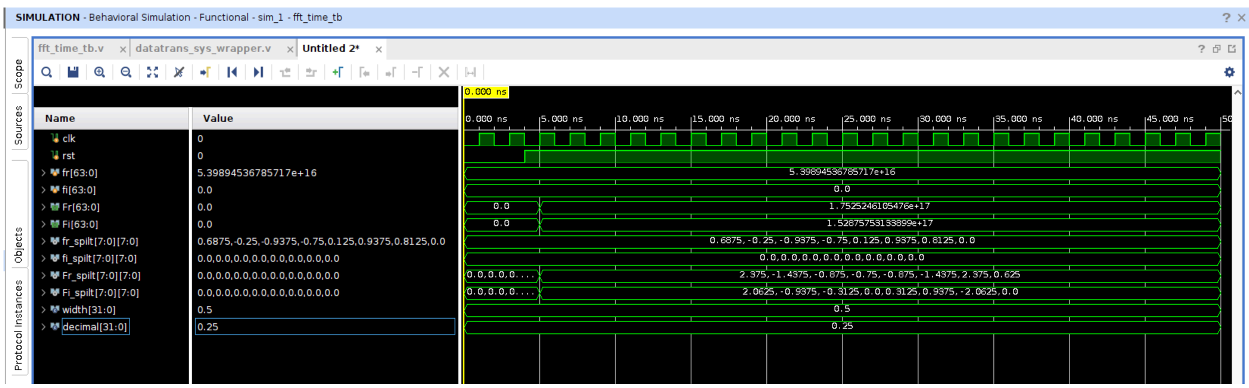
1. Open the provided lab4_base Vivado project and add sources: fft_time.v, butterfly.v, multiply.v, and do not modify fft_time_tb.v. Use 8-bit signed fixed-point with 4 fractional bits.
2. In the testbench, fr/fi are input real/imag parts and Fr/Fi are output real/imag. Set the waveform radix for the split buses to floating-point to read values easily.
3. Make fft_time_tb the simulation top, set the sim length (≈ 50 ns is sufficient), and run Run Behavioral Simulation.

Results:

1. After rst de-asserts, the 8 real input samples (Fr_split_r) are applied in Q4.4 and the imaginary inputs are 0, as intended. After a short pipeline fill (three butterfly stages for an 8-point radix-2 DIT), the outputs Ff_split_r (real) and Ff_split_i (imag) populate with 8 complex values each.
2. The output vectors are stable (no X/Z), change only on clock edges, and remain constant once produced. The imaginary part is ~0 at DC/Nyquist, and the spectrum shows the expected conjugate symmetry for a real-valued time sequence. Quantized values track the fixed-point format (Q4.4) used by the design.







Inference:

- The timing of output availability matches the 3-stage DIT pipeline, confirming correct stage-by-stage propagation through the butterflies.
- The numeric pattern of Ff_split_r/i is consistent with the FFT of the shown input: DC and harmonic bins have plausible magnitudes/signs, and small discrepancies are within 1 LSB of Q4.4 quantization/rounding.
- Signal integrity is good (clean reset, synchronous updates), indicating that the testbench–design interface and radix settings are correct.

Conclusion:

The behavioral simulation verifies functional correctness of the 8-point radix-2 DIT FFT: outputs appear after the expected pipeline latency, exhibit the correct complex spectrum for a real input, and remain stable thereafter. The design is ready to proceed to synthesis/implementation, with any further scaling or ordering (natural vs. bit-reversed) handled per the testbench's mapping.